

Securing Wireless Communications via Channel Reciprocity and Dynamic Constellation Obfuscation

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Abstract—The one-time pad secure transmission based on wireless channel reciprocity (CR-OTP) has drawn great attention recently due to its capability of providing perfect secrecy of data, as well as the modulation information. However, existing CR-OTP schemes encounter both reliability and security challenges as their assumptions of channel reciprocity and randomness are not always well satisfied in practical application scenarios. To tackle these issues, we propose a dynamic constellation obfuscation (DCO) method that obfuscates the plaintext by rotating its constellation dynamically. This kind of analog encryption method is proven to be more robust than the existing digital exclusive OR (xor) encryption method as the former achieves a lower symbol error rate (SER) by reducing the double quantization loss to one. The rotation pattern is jointly dependent on the channel state information (CSI) and the previous message, which guarantees the randomness of the rotation pattern subjected to environmental drifts. Only the legitimate receiver that correctly recovers the previous message correctly and observes a similar CSI is able to decode the newly transmitted message. We proved that the secrecy capacity of the proposed DCO method is higher than that of the state-of-the-art. Simulation results confirm that the proposed method delivers superior performance regarding secrecy capacity and SER, achieving a 4.5 dB signal-to-noise ratio (SNR) gain at a SER of 0.1; moreover, when the secrecy capacity is 0.1, the main channel SNR gain reaches 6.5 dB when the wiretap channel SNR is 20 dB.

Index Terms—Constellation obfuscation, constellation rotation, differential attack, physical-layer encryption (PLE), rotation patterns.

I. INTRODUCTION

THE EVER-INCREASING demand for private and sensitive data transmission over wireless networks has made

security a crucial concern in the current and future large-scale, dynamic, and heterogeneous wireless communication systems [1]. In existing communication systems, the issue of data confidentiality is typically handled in the upper layers of the protocol stack using a variety of cryptography algorithms [2]. While the relevant cryptosystems, even with unaltered root key, are proven secure for an attacker with limited computational power, they might be rendered insecure by the rapid progress of computing techniques in the quantum era [3]. Moreover, state-of-the-art cryptosystems require a trusted third party for key distribution and are typically computationally intensive [4], [5], [6]. Consequently, their application in machine-to-machine communications or latency-sensitive applications could be challenging [7]. Therefore, it becomes urgent to find a lightweight solution for securing wireless transmission with high-level security.

In recent years, wireless researchers have sought a new security paradigm, colloquially known as channel reciprocity-based one-time pad secure transmission (CR-OTP) [8]. CR-OTP aims to reinforce the security of communication systems against quantum computers by leveraging the shared entropy source of wireless fading channels. Because of the inherent reciprocity principle of wireless channels, two legitimate parties, namely, Alice and Bob, will obtain statistically dependent channel state information (CSI) as long as they observe their mutual channel at approximately the same time. For this reason, Alice and Bob can generate secret keys from their unique CSI and use them as one-time random keys for secure transmission [9]. This kind of random key can be used to provide protection not only for the transmitted data but also for the control and header information, thereby avoiding traffic analysis attacks, such as international mobile subscriber identity (IMSI) catcher [10]. When the surrounding environment is time-varying, the secret key will be updated, and the location-specific feature of CSI will prevent any eavesdroppers located several wavelengths away from knowing the key or data [11].

Despite above advantages of CR-OTP, this technique is still not mature to be implemented in existing communication systems. The underlying challenges come from two aspects, i.e., reliability and security. First, the similarity of the CSI estimates at Alice and Bob is susceptible to the half-duplex mode, hardware mismatches, and noise, potentially resulting in high disagreement between their quantized bit sequences. In this context, the key generation might be unsuccessful even with information reconciliation, which significantly affects the effectiveness of CR-OTP. Moreover, the security concern about

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CR-OTP comes from its heavy dependency on the channel variation. However, the dynamic change of wireless channels is uncontrollable, meaning that there always exists more or less temporal correlations among adjacent channel samples. For this reason, CR-OTP suffers from vulnerabilities under attacks, especially in slowly varying environments [12]. For example, the correlation of adjacent frames can be used by an eavesdropper to conduct a differential attack, which will be discussed in the sequel. Although there have been ongoing research efforts on improving the dynamic property of channel parameters by the reconfigurable intelligent surface (RIS) [13], these schemes introduce additional hardware overheads, and their effects are somewhat limited considering the cascaded path loss of the RIS-induced path. Therefore, a reliable and secure solution is still required to facilitate the practical usage of CR-OTP.

In this regard, this article designs a new secure wireless communication approach based on the Dynamic Constellation Obfuscation (DCO) technique that integrates channel reciprocity and analog encryption for the enhancement of transmission reliability and security. In our proposed approach, Alice obfuscates the transmitted message by rotating its constellation according to a fast-changing rotating pattern that is generated by using the previous message and the current extracted CSI. Bob then decrypts the message upon reception, converting it from ciphertext back to plaintext by rotating the constellation in a reversed way with its own CSI and previous message. Unlike digital exclusive OR (XOR) encryption, DCO does not need to quantize CSI and thus avoids the double quantization loss in existing CR-OTP methods. In this respect, DCO is expected to achieve better reliability for secure communications. In DCO, the time-varying rotation pattern is jointly decided by the CSI and the previous message. This idea is inspired by the finding that just like wireless channels, transmitted messages also exhibit certain random characteristics. The randomness of message comes from the unpredictability of the information source, which is generally modeled as a stochastic process.

To verify this hypothesis, we carried out a preliminary experiment by calculating the entropy of messages chosen from two classical image datasets, i.e., COCO and MNIST, with two typical source coding methods, i.e., Huffman encoding and run-length encoding (RLE). As observed from Fig. 1, for all kind of messages, their probabilities of entropy are concentrated in the interval from 0.85 to 1, indicating the good randomness of message.

Taking advantage of this observation, DCO uses previous message that has already been shared with Alice and Bob as another source of randomness for generating time-varying rotation pattern. Only the receivers that correctly recover the previous message and obtain similar CSI are able to decode the currently transmitted message. In this way, DCO provides an enhancement mechanism for communication security in slowly-varying environments.

The main contributions of this article are listed as follows.

- 1) We propose a new secure wireless communication approach referred to as DCO, which reduces the double quantization loss of CR-OTP to one. The proposed

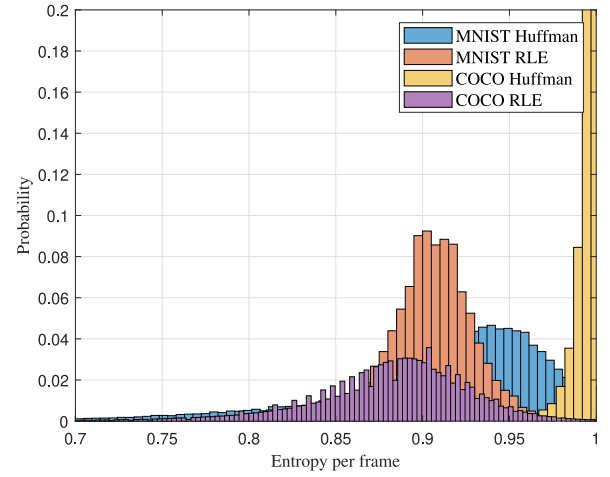


Fig. 1. Probability distribution of the entropy of the message for different datasets and different source coding methods, which are extracted from 60 000 images from two public datasets, i.e., COCO and MNIST.

approach is proven to have a lower symbol error rate (SER) than the existing digital XOR-based PLE method and thus achieves higher reliability.

- 2) We propose a time-varying rotation pattern generation method by jointly using the shared randomness extracted from the channel and previous messages. The new design enhances the security of DCO in slow-varying environments. The secrecy capacity of the proposed DCO method is proven to be higher than the state-of-the-art PLE schemes, which mostly ignore the natural randomness in the data.
- 3) We verify the performance of the proposed methods via numerical simulations. The obtained simulation results show that the DCO is able to significantly improve the secrecy capability and SER performance compared with the conventional XOR scheme, with an SNR gain of 4.5 dB when the SER is 0.1 and 5 dB when the secrecy capability is 0.1.

In this article, we analyze the SER and security performance of the XOR and DCO schemes while extending the DCO scheme. Furthermore, we demonstrate from theoretical and simulation results that the proposed DCO scheme has higher secrecy capacity and lower SER.¹

We use the following notations throughout the article. upper (lower) bold-face letters denote matrices (column vectors); $\mathbf{I}$ denotes the identity matrix while its subscript, if needed, represents its dimensionality. $\mathbb{C}^{M \times N}$ denotes the space of complex matrices of size $M \times N$. Superscripts $(\cdot)^H$, $(\cdot)^T$, and $(\cdot)^*$ stand for the conjugate-transpose, transpose, and conjugate of the enclosed matrix, respectively.

II. RELATED WORK

DCO builds on past works on CR-OTP as well as signal constellation mapping-based physical-layer encryption

¹The content in this article has been partially published by IEEE VTC FALL 2022. In our previous work, we focused on dynamic constellation rotation and proposed a scheme to dynamically update the constellation symbols that can reduce the pilot signaling overhead and increase the randomness of the constellation.

(PLE). Li et al. established the mathematical models, design frameworks, and cryptographic primitives of PLE in [8]. Rather than the bit-level encryption used in CR-OTP, PLE achieves encryption via subcarrier scrambling [14], frequency-shift mapping [15], constellation scrambling [16], directional modulation [17], and artificial noise [18], etc. For example, Nain et al. proposed a phase encryption method for various modulation schemes and applied it to the application scenarios of IEEE 802.15.4 via FPGA and ASIC implementation [19]. In [20], Xiong et al. adopted a multiple intersymbol obfuscation (MIO) scheme to enhance wireless communications security at the physical layer. In MIO, upon sending each data packet, a random subset of the corresponding data symbols is obfuscated with a set of artificial noisy symbols, which is called the symbols key. Compared with the previous constellation design approaches, the constellation diagram of the MIO encryption scheme is more difficult to crack. The primary advantage of these PLE strategies lies in their ability to create unique signal constellation mapping techniques, which effectively prevent eavesdroppers from correctly mapping received digital symbols into bits, thereby leading to erroneous decoding of packets. For this reason, PLE schemes can enhance the security level dramatically and resist various attacks, such as cryptanalysis and traffic analysis attacks [8], [21]. PLE schemes also improve the lifetime of the node in the presence of a ghost attacker by preventing the legitimate node from processing the bogus messages and hence combat energy depletion attacks [19].

However, similar to conventional symmetric key encryption, PLE still requires a preshared secret key [19], [20], while previous works rarely considered the key distribution issue. When the key is constant, the resulted security is hardly guaranteed as an attacker could accurately identify the rotation information using well-known attack strategies, such as adaptive plaintext attacks and brute-force search attacks. Although chaotic systems can be incorporated to enlarge key space and enhance key sensitivity [22], [23], [24], its security will be compromised as long as the initial values are leaked. The inherent characteristics of wireless channels make it a good candidate as a shared source of randomness. In this context, Hu et al. proposed a modulated symbols-based one-time pad (SOTP) secure transmission scheme which exploits physical-layer keys generated from channel observations for symbols encryption [25]. Although SOTP made a tentative exploration on combining CR-OTP and PLE, its assumption of the random channel is idealized and oversimplified. In particular, SOTP did not consider the weak channel randomness in practical slow-varying environments and the consequent vulnerabilities under various attacks [12].

Distinct from previous works, the DCO scheme proposed in this article obfuscates constellation symbols by jointly utilizing the randomness from both wireless channels and messages, without the need for trusted third parties, preshared sequences, or particular channel conditions. Moreover, the proposed DCO scheme can implement constellation rotation encryption using analog values, providing increased security and dependability in various lightweight communication settings.

III. SYSTEM MODEL AND PROBLEM STATEMENT

This section introduces the system model and on this basis describes the problem aimed to be solved in this article.

A. System Model

We consider a classical communication system that involves three communication parties that operate in a time-division duplex (TDD) mode. For simplicity, all parties are assumed to be equipped with a single antenna. Alice and Bob form the legitimate communication pair, which communicate confidential information exclusively between each other. Meanwhile, Eve, an eavesdropper, attempts to intercept the confidential information communicated between Alice and Bob. Physically, Eve can only approach one end, and, without loss of generality, we assume Eve to be close to Bob, but at least several wavelengths away for avoiding being noticed.

Since Alice and Bob aim at sampling their channel randomness for DCO, we start with modeling channel sampling process along with the wireless channels between them. Specifically, at the k th sampling instant, denoted as $t_a(k)$, Alice sends a request packet with orthogonal frequency division multiplexing (OFDM) symbols to Bob, who will obtain measurement signal $\mathbf{y}_B(k)$ as

$$\mathbf{y}_B(k) = \mathbf{S}_A(k)\mathbf{h}_{AB}(k) + \mathbf{w}_B(k). \quad (1)$$

After a short time delay Δ_t , Bob replies with a packet to Alice, who will also measure the same parameter and obtain signal $\mathbf{y}_A(k)$ as

$$\mathbf{y}_A(k) = \mathbf{S}_B(k)\mathbf{h}_{BA}(k) + \mathbf{w}_A(k) \quad (2)$$

where $\mathbf{h}_{AB}(k) \in \mathbb{C}^{N \times 1}$ and $\mathbf{h}_{BA}(k) \in \mathbb{C}^{N \times 1}$ are the channel responses from Alice to Bob and from Bob to Alice, respectively; N is the number of subcarriers; $\mathbf{S}_u(k) = \text{diag}(s_1(k), s_2(k), \dots, s_N(k))$ represents the pilot symbol from node $u \in \{A, B\}$. The pilot symbol is subject to the average power constraint as

$$\frac{1}{N} \sum_{i=1}^N \left\{ |s_i(k)|^2 \right\} = P. \quad (3)$$

In addition, $\mathbf{w}_u(k) \in \mathbb{C}^{N \times 1}$ is the complex additive white Gaussian noise (AWGN) vector with variance σ_n^2 at node u .

Because in TDD schemes, data transmissions in both directions use the same carrier frequency, unless strong frequency-selective fading and different co-channel interference are encountered, the complex-valued channel envelope remains near-constant during coherence time T_c . As long as $\Delta_t < T_c$, we can assume perfect channel reciprocity and denote

$$\mathbf{h}_{BA}(k) = \mathbf{h}_{AB}(k) = \mathbf{h}_m(k) \quad (4)$$

where $\mathbf{h}_m(k)$ stands for the coefficient vector of the main channel, and each element in $\mathbf{h}_m(k)$ is Gaussian distributed with zero mean and the variance of σ_m^2 . Based on the received signal, Alice and Bob are then able to estimate their channel

using the least square (LS) method as

$$\hat{\mathbf{h}}_A(k) = \mathbf{S}_B(k)^{-1} \mathbf{y}_A(k) = \mathbf{h}_m(k) + \mathbf{z}_A(k) \quad (5)$$

and

$$\hat{\mathbf{h}}_B(k) = \mathbf{S}_A(k)^{-1} \mathbf{y}_B(k) = \mathbf{h}_m(k) + \mathbf{z}_B(k) \quad (6)$$

where channel estimation errors $\mathbf{z}_A(k) = \mathbf{S}_B(k)^{-1} \mathbf{w}_A(k)$ and $\mathbf{z}_B(k) = \mathbf{S}_A(k)^{-1} \mathbf{w}_B(k)$ are assumed to be Gaussian distributed with zero mean and variance σ_n^2/P , i.e., $\mathbf{z}_A, \mathbf{z}_B \sim \mathcal{CN}(0, \sigma_n^2/P)$. The signal-to-noise ratio (SNR) of the main channel is thus given by

$$r_m = \frac{P \|\mathbf{h}_m(k)\|_2^2}{\|\mathbf{w}_m(k)\|_2^2} = \frac{P \sigma_m^2}{\sigma_n^2}. \quad (7)$$

Then, Alice and Bob will repeat the above sampling for every time interval of Δ_t .

Similarly, Eve is also capable of eavesdropping on Alice's transmissions, which are given by

$$\mathbf{y}_E(k) = \mathbf{S}_A(k) \mathbf{h}_w(k) + \mathbf{w}_E(k) \quad (8)$$

where $\mathbf{h}_w(k) \sim \mathcal{CN}(0, \sigma_w^2)$ represent the wiretap channel vector, and $\mathbf{w}_w(k) \in \mathbb{C}^{N \times 1}$ is the complex AWGN vector. Based on the received signal, Eve can estimate the wiretap channel via

$$\hat{\mathbf{h}}_E(k) = \mathbf{S}_A(k)^{-1} \mathbf{y}_E(k) = \mathbf{h}_w(k) + \mathbf{z}_E(k) \quad (9)$$

where $\mathbf{z}_E(k) = \mathbf{S}_A(k)^{-1} \mathbf{w}_E(k) \sim \mathcal{CN}(0, \sigma_n^2/P)$. When the distance between Eve and the legitimate user exceeds half the signal wavelength $\lambda/2$, the eavesdropper's channel becomes uncorrelated with that of the legitimate user [26].

The SNR of the wiretap channel can thus be expressed as

$$r_w = \frac{P \|\mathbf{h}_w(k)\|_2^2}{\|\mathbf{w}_E(k)\|_2^2} = \frac{P \sigma_w^2}{\sigma_n^2}. \quad (10)$$

B. Threat Model

We consider that Eve's goal is to infer confidential information between Alice and Bob passively through monitoring and analyzing the collected ambient wireless signals. Given this setup, we summarize potential attacks into the following five categories.

- 1) *Statistical/Traffic Analysis Attack*: Eve tries to obtain the modulation information by statistically analyzing the signal constellation of received OFDM symbols. To achieve this purpose, Eve can exploit statistical information analysis methods, such as likelihood ratio-based verdicts and pattern-based identification [27].
- 2) *Symbol Differential Attack*: If the obfuscation parameters of adjacent messages are invariant over a long period of time, Eve can eliminate the obfuscation and learn the differential information of these confidential messages through reverse operations, e.g., conjugate multiplication.
- 3) *Injection Attack*: During the initial channel probing phase using pilot signals by the legitimate parties, Eve injects the same channel phase into both Alice and Bob

simultaneously to manipulate their channel estimation results [28].

- 4) *Channel Replay Attack*: Eve tries to obtain a similar channel signature as that of Bob by recording Bob's location and moving to it after Bob moves. On this basis, Eve can decouple the constellation pattern and eliminate the randomness caused by wireless channels [29].
- 5) *Channel Manipulating Attack*: Eve tries to control the randomness of the channel environment between Alice and Bob, for example, by arranging some wave-absorbing environments to make the current channel have only line-of-sight (LoS) paths, which leads to slow channel variation and highly correlated coefficients among channels for conveying neighboring frames.
- 6) *Message Manipulating Attack*: Eve tries to infer some modulation information or channel information by inducing Alice to send repeated or low entropy messages. This attack can be used to eliminate the randomness caused by messages.

It is worth noting that the statistical/traffic analysis attack and symbol differential attack can be combined with the channel replay attack, channel manipulating attack, and message manipulating attack to launch in practice [30].

C. Problem Statement

Since Alice and Bob are expected to observe highly correlated channel estimates, i.e., $\hat{\mathbf{h}}_A(k)$ and $\hat{\mathbf{h}}_B(k)$, they intend to exploit them for secure communications. An intuitive method involves quantifying these channel estimates to raw keys $\mathbf{q}_A(k)$ and $\mathbf{q}_B(k)$ first and then employing them for one-time pad encryption through either identical or unidentical ways, as incorporated in CR-OTP [8]. However, this method encounters both reliability and security challenges in practical usage.

- 1) *Reliability*: As observed, $\hat{\mathbf{h}}_A(k)$ and $\hat{\mathbf{h}}_B(k)$ are not completely consistent. Moreover, quantization, for converting analog measurement estimates into discrete binary sequences, will inevitably result in quantization loss and thus deteriorate the inconsistency between $\mathbf{q}_A(k)$ and $\mathbf{q}_B(k)$. Although these deviations are expected to be corrected through information reconciliation, this process becomes inefficient when these deviations go large, due to the limited error-correction capability and a high-percentage of information leakage. For this reason, we should exhaust all means to improve the agreement between $\mathbf{q}_A(k)$ and $\mathbf{q}_B(k)$ from the very beginning.
- 2) *Security*: In CR-OTP, messages are encrypted before modulation, therefore the modulation information is not stealthy and vulnerable to traffic analysis attacks. Moreover, when sampling interval Δ_t is smaller than channel coherence time T_c , the channels of adjacent moments are inevitably relevant. If these channel estimates are used for encryption directly, their temporal correlation would compromise the security level. Although existing works tried to avoid correlated samples by magnifying Δ_t of over T_c , this solution is unrealistic due to the difficulties of obtaining an accurate estimation of T_c and a timely adjustment of Δ_t .

Therefore, the core problem to be solved in this article is securing wireless communications via channel reciprocity while addressing the aforementioned reliability and security issues.

IV. SECURE COMMUNICATION SCHEME BASED ON DYNAMIC CONSTELLATION OBFUSCATION

This section introduces the framework of secure communications based on DCO, with an emphasis on the design of two core modules for constellation pattern generation and constellation obfuscation.

A. Framework Description

Fig. 2 depicts the framework of DCO-based secure communications, which integrates channel reciprocity and DCO for the enhancement of reliability and security. For current message X , Alice obfuscates it by rotating its constellation according to the constellation pattern generation module, which yields a fast-changing pattern using the delayed message context and the current extracted CSI, denoted as CSI_A . The transmitted and received obfuscated signals are denoted by X^c and $\hat{X}^c$, respectively. Bob then decrypts the message upon reception, converting from ciphertext $\hat{X}^c$ back to plaintext $\hat{X}$ by rotating the constellation in a reversed way with its own CSI, denoted as CSI_B , to obtain the delayed decrypted message.

The design of constellation pattern pursues both aspects of similarity and dynamics. Only when the constellation patterns of Alice and Bob are similar, they are able to achieve the agreement of decrypted messages with a high probability. Moreover, the dynamics of constellation pattern prevents Eve from learning the regularity of obfuscation mode to crack.

To achieve this goal, as shown in Fig. 3, the generation of the constellation pattern is divided into two stages: the former involves generating a pair of similar initial constellation patterns, while the latter focuses on updating these constellation patterns dynamically. We generally denote $\mathbf{e}(k)$ as the constellation pattern of the k th symbol, which satisfies

$$\mathbf{e}(k) = \begin{cases} \Phi, & k = 0, \\ \mathbf{d}(k), & 0 < k \leq M \end{cases} \quad (11)$$

where Φ and $\mathbf{d}(k)$ are the initial constellation pattern and the updated constellation pattern of the k th symbol over all subcarriers, respectively. The specific designs of both for each subcarrier are elaborated as follows.

1) *Initial Constellation Pattern*: Although the wireless channel itself can provide the initial shared randomness, the training symbol sent in plaintext will permit Eve to easily collect the modulation information. In this context, we exploit the technique of private pilots as discussed in [31] to address this problem.

Define the initial constellation pattern as $\Phi = [\Phi_1, \Phi_2, \dots, \Phi_N]$, where Φ_i is the pattern over the i th subcarrier. To hide the modulation information, we exploit the random private pilot to generate Φ_i as follows. As shown in Fig. 3(a), Alice and Bob initialize the constellation pattern by sending random private pilots to each other alternatively.

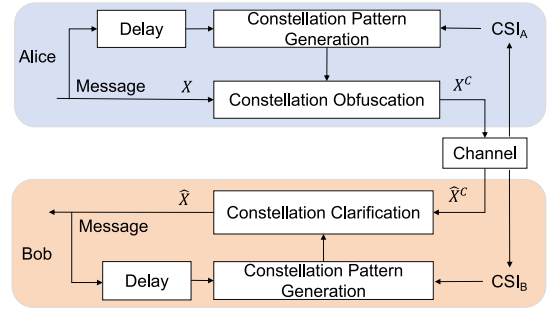


Fig. 2. Framework of the secure communications approach based on DCO.

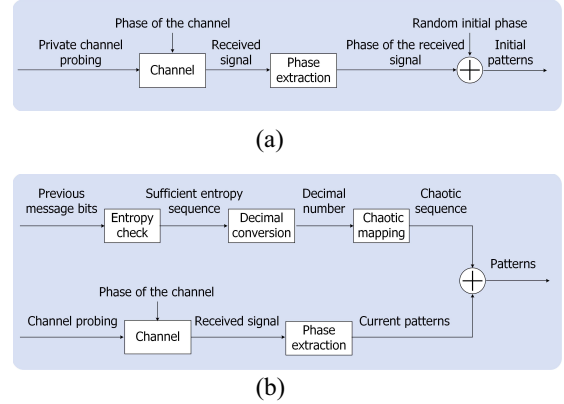


Fig. 3. Framework of constellation pattern generation for secure communications. (a) Initial constellation pattern generation. (b) Constellation pattern update.

First, Alice generates and transmits the private probing signal at $k = 0$ as

$$\mathbf{S}_A(0) = \text{diag}(e^{j\varphi_{A,1}}, e^{j\varphi_{A,2}}, \dots, e^{j\varphi_{A,N}}) \quad (12)$$

where $\varphi_{A,i}$ is the initial phase randomly generated on a uniform basis over $[0, 2\pi)$ for the i th OFDM subcarrier. Then, its corresponding signal received by Bob can be written as

$$y_{B,i}(0) = |h_{m,i}(0)|e^{j(\varphi_{A,i} + \theta_{m,i}(0))} + w_{B,i}(0) \quad (13)$$

where $|h_{m,i}(0)|$ and $\theta_{m,i}(0)$ denote the amplitude and phase of the main channel for conveying the i th OFDM subcarrier, respectively; $w_{B,i}(0)$ denotes the AWGN at Bob.

Once receiving the downlink probing signal, Bob sends its private probing signal at $k = 0$ as

$$\mathbf{S}_B(0) = \text{diag}(e^{j\varphi_{B,1}}, e^{j\varphi_{B,2}}, \dots, e^{j\varphi_{B,N}}) \quad (14)$$

to Alice, where $\varphi_{B,i}$ is the initial phase randomly generated on a uniform basis over $[0, 2\pi)$ for the i th OFDM subcarrier. Due to the channel reciprocity, the corresponding signal received by Alice can be written as

$$y_{A,i}(0) = |h_{m,i}(0)|e^{j(\varphi_{B,i} + \theta_{m,i}(0))} + w_{A,i}(0) \quad (15)$$

where $w_{A,i}(0)$ denotes the additive Gaussian noise at Alice.

According to the received signal, Alice and Bob construct their initial patterns by modulo-adding the phase of the received signal with that of the private pilot as

$$\Phi_{A,i} = (\Phi(y_{A,i}(0)) + \varphi_{A,i}) \bmod 2\pi \quad (16)$$

and

$$\Phi_{B,i} = (\Phi(y_{B,i}(0)) + \varphi_{B,i}) \bmod 2\pi \quad (17)$$

respectively. Here, $\Phi(\cdot)$ represents the phase of the enclosed complex argument. By substituting (15) and (13) into (16) and (17), similar initial patterns can be achieved as

$$\Phi_{A,i} \approx \Phi_{B,i} \approx \Phi_i = (\varphi_{A,i} + \theta_{m,i}(0) + \varphi_{B,i}) \bmod 2\pi. \quad (18)$$

Considering the influence of nonreciprocal factors, such as rapid environmental changes or CSI estimation errors, $\Phi_{A,i}$ and $\Phi_{B,i}$ are not exactly the same, and we define

$$\varepsilon_i = |\Phi_{A,i} - \Phi_{B,i}| \quad (19)$$

as their absolute deviation. We can model ε_i as a uniformly distributed random variable in the interval $[-\Delta, +\Delta]$, and Δ is the maximum phase error of the channel [32].

2) *Updated Constellation Pattern*: Although private pilots can be employed to hide the modulation information, their communication overhead becomes barely affordable when we use this technique frequently for pattern update. For this reason, we update the constellation pattern dynamically by using both previous messages and the current extracted CSI as

$$\mathbf{d}(k) = \boldsymbol{\theta}(k) + \mathbf{c}(k) \quad (20)$$

where $\boldsymbol{\theta}(k)$ and $\mathbf{c}(k)$ are the pattern of the channel and that constructed through previous messages from the k th symbol. For the i th subcarrier, $\theta_i(k)$ is obtained by

$$\theta_i(k) = 2\pi \frac{\Re(\hat{h}_{m,i}(k))}{\max(\Re(\hat{h}_m(k)))}, k \geq 1 \quad (21)$$

where $\max(\hat{h}_m(k))$ represents the maximal value of the real and imaginary parts of $\hat{h}_{m,i}(k)$ over all subcarriers. In this article, we take the real part as an example, and the same goes for the imaginary part.

Fig. 3(b) illustrates the strategy of pattern update. To guarantee sufficient randomness, we select multiple previous messages using a sliding window with length of w . Let $\mathbf{b}_i(k, w) = [b_i(k-1), b_i(k-2), \dots, b_i(k-w)]$ denote the selected binary sequence of the k th symbol over the i th subcarrier. Window size w , i.e., the number of used messages, is decided according to the required entropy of the sequence given by

$$\mathbf{H}(\mathbf{b}_i(k, w)) \geq H_\tau \quad (22)$$

where $\mathbf{H}(\mathbf{b}_i(k, w))$ denotes the entropy of the sequence $\mathbf{b}_i(k, w)$, and H_τ is a preset threshold of entropy. Under the assumption of independent messages, we can employ the binary search method to efficiently select the minimum value of w in order to construct a sliding window as $[k-w, k-1]$.

The sliding window dynamically adjusts its length w based on the entropy of the message sequence $\mathbf{H}(\mathbf{b}_i(k, w))$, ensuring $\mathbf{H}(\mathbf{b}_i(k, w)) \geq H_\tau$. This guarantees sufficient randomness even for repetitive data.

For some extreme low-entropy scenarios (e.g., static data streams), hybrid entropy sources (e.g., timestamps or

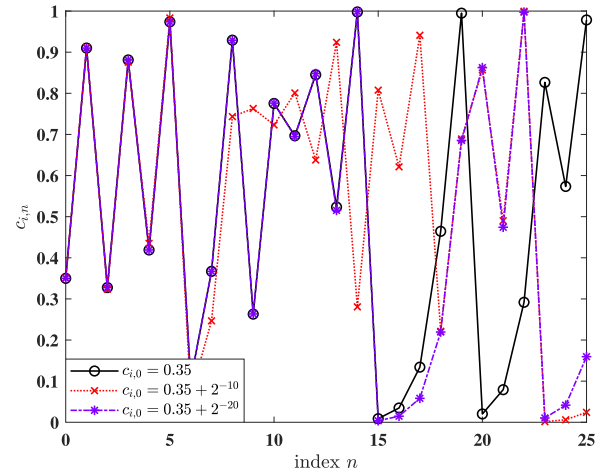


Fig. 4. Comparison of $c_{i,n}$ with different initial values $c_{i,0}$.

hardware-generated random numbers) can be integrated into the chaotic system's initial value, reducing dependence on message entropy alone

$$\mathbf{b}_i^{enc}(k, w) = \mathbf{b}_i(k, w) \oplus \text{RNG}_{hw}(k, w), \quad (23)$$

where $\text{RNG}_{hw}(k, w)$ denotes a Hardware-based Random Number Generator with k, w as parameters. Then, message sequence $\mathbf{b}_i^{enc}(k, w)$ is transformed into a decimal number $b_{i,dec}(k)$ through

$$b_{i,dec}(k) = \frac{1}{w} \sum_{j=0}^{w-1} b_i^{enc}(k-j-1) \times 2^{-j}. \quad (24)$$

To enhance the security, we use $b_{i,dec}(k)$ as the initial value $c_{i,0}$ of the chaotic sequence as a chaotic system can generate key sequences with special chaotic characteristics, such as pseudo randomness, ergodicity, and high sensitivity to the initial values [33].

To facilitate the following explanation and mathematical analysis, we utilize the logistic map given by

$$c_{i,n+1} = 4c_{i,n}(1 - c_{i,n}). \quad (25)$$

When the initial value satisfies $0 < c_{i,0} < 1$, the logistic mapping is considered to be in a chaotic state.

As an illustrative example, Fig. 4 displays the transition of $c_{i,n}$ ($n = 0, 1, \dots, 25$) with initial values $c_{i,0} = 0.35$, $0.35 + 2^{-10}$, and $0.35 + 2^{-20}$. Despite the minuscule differences among these initial values, resulting $c_{i,n}$ exhibits significant divergence after multiple iterations. This observation emphasizes that the precision of the initial value for iteration dominates the length of the chaotic sequence in consistency. Consequently, for realizing secure communications, one must carefully initialize the chaotic system iterations based on the desired sensitivity level for encrypted information. Without loss of generality, we take $\mathbf{b}_{dec}(k)$ and iterate I times to generate a chaotic sequence $\mathbf{c}(k) \in \mathbb{C}^{N \times 1}$, which is then mapped linearly to interval $[0, 2\pi)$ as follows:

$$\mathbf{c}(k) = 2\pi [c_{1,I}(k), c_{2,I}(k), \dots, c_{N,I}(k)]. \quad (26)$$

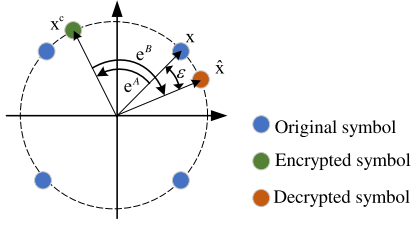


Fig. 5. Constellation rotation and reverse rotation for QPSK.

Substituting (26) into (20), it can be observed that even though initial phase Φ remains constant, the difference of transmitted data per frame can still vary $\mathbf{c}(k)$, resulting in $\mathbf{e}(k)$ to be highly dynamic.

B. Constellation Obfuscation and Clarification

To facilitate accurate recovery of received bits at the legitimate end, we propose an encryption and decryption approach based on random constellation rotation, referred to as constellation *obfuscation* and *clarification*, respectively. The DCO encryption and decryption algorithm is shown in Algorithm 1.

Specifically, bit sequence $\mathbf{b}(k)$ at Alice is mapped to N parallel modulation symbols from the symbol constellation, with the k th symbols represented as $\mathbf{x}(k)$, where $k = 0, 1, \dots, M$. Following serial-to-parallel conversion, standard modulation symbols undergo encryption via constellation rotation:

$$\mathbf{x}^c(k) = \mathbf{E}^A(k)\mathbf{x}(k) \quad (27)$$

where $\mathbf{E}^A(k) = \text{diag}(e^{je_1^A(k)}, e^{je_2^A(k)}, \dots, e^{je_N^A(k)})$ denotes a phase rotation matrix generated according to (11), and $\mathbf{x}^c(k) = [x_1^c(k), x_2^c(k), \dots, x_N^c(k)]^T$ represents the encrypted OFDM symbols. Subsequently, $\mathbf{x}^c(k)$ is transformed into time-domain symbols through inverse fast Fourier transform (IFFT), which is loaded on the subcarrier and transmitted after parallel-to-serial conversion, cyclic prefix (CP) appending, and pilot insertion.

As shown in Fig. 2, Bob carries out the reverse procedure of Alice's operations to obtain the encrypted OFDM symbols and perform inverse rotation decryption to retrieve the modulated symbols as follows:

$$\hat{\mathbf{x}}(k) = \mathbf{E}^B(k)\mathbf{x}^c(k) \quad (28)$$

where $\mathbf{E}^B(k) = \text{diag}(e^{-je_1^B(k)}, e^{-je_2^B(k)}, \dots, e^{-je_N^B(k)})$ represents a phase inverse rotation matrix.

The constellation rotation and inverse rotation under quadrature phase shift keying (QPSK) are illustrated in Fig. 5. For $i \in 1, 2, \dots, N$, when $e_i^A(k)$ and $e_i^B(k)$ belong to distinct quantization intervals, their phase error is not sufficient to alter the symbol quadrant, but our scheme can still be decrypted correctly. In contrast, by the traditional digital XOR encryption method, $e_i^A(k)$ and $e_i^B(k)$ are quantized into keys directly, and therefore inconsistent keys will be generated, resulting in decryption errors.

Algorithm 1 Encryption and Decryption Algorithm Based on DCO

Require:

M data frame bits are to be transmitted.

Φ^A is generated at Pattern Initialization stage.

Ensure:

$\mathbf{b}(k) \rightarrow \mathbf{x}(k) \rightarrow \mathbf{x}^c(k) \rightarrow \hat{\mathbf{x}}(k) \rightarrow \hat{\mathbf{b}}(k)$

- 1: // **Initialization**
- 2: Alice sends a private pilot $\mathbf{S}_A(0)$ and stores it locally;
- 3: Bob sends a private pilot $\mathbf{S}_B(0)$ and stores it locally;
- 4: Combine the locally stored random pilots and estimated signal phase to generate Φ^A and Φ^B ;
- 5: // **Alice: Encryption Process**
- 6: **for** $k = 0$ to M **do**
- 7: **if** $k = 0$ **then**
- 8: Initialize $\mathbf{e}^A(k)$ as Φ^A ;
- 9: **else**
- 10: Construct the sliding window $[k - w, k - 1]$ based on the entropy evaluation;
- 11: Convert the bits in the sliding interval to $\mathbf{b}_{dec}^A(k)$ with RNG_{hw} ;
- 12: Using the $\mathbf{b}_{dec}^A(k)$ as input of the chaotic function to generate $\mathbf{c}^A(k)$;
- 13: Update constellation patterns $\mathbf{e}^A(k) = \mathbf{c}^A(k) + \theta^A(k)$;
- 14: **end if**
- 15: Map k th frame bits $\mathbf{b}(k)$ to N data symbols $\mathbf{x}(k)$;
- 16: **for** $i = 1$ to N **do**
- 17: $\mathbf{x}_i^c(k) = \mathbf{x}_i(k) \exp\{je_i^A(k)\}$;
- 18: **end for**
- 19: **end for**
- 20: // **Bob: Decryption Process**
- 21: **for** $k = 0$ to M **do**
- 22: **if** $k = 0$ **then**
- 23: Initialize $\mathbf{e}^B(k)$ as Φ^B ;
- 24: **else**
- 25: Construct the sliding window $[k - w, k - 1]$ based on the entropy evaluation;
- 26: Convert the bits in the sliding interval to $\mathbf{b}_{dec}^B(k)$ with RNG_{hw} ;
- 27: Using the $\mathbf{b}_{dec}^B(k)$ as input of the chaotic function to generate $\mathbf{c}^B(k)$;
- 28: Update constellation patterns $\mathbf{e}^B(k) = \mathbf{c}^B(k) + \theta^B(k)$;
- 29: **end if**
- 30: **for** $i = 1$ to N **do**
- 31: $\hat{\mathbf{x}}_i(k) = \mathbf{x}_i^c(k) \exp\{-je_i^B(k)\}$;
- 32: **end for**
- 33: **end for**
- 34: Map received symbols $\hat{\mathbf{x}}(k)$ to bits $\hat{\mathbf{b}}(k)$;

V. PERFORMANCE ANALYSIS

In this section, we first give the analysis of the robustness of the proposed DCO scheme and subsequently perform a security analysis for the system employing the DCO scheme.

A. Symbol Error Rate Analysis

To demonstrate the advantages of DCO over the existing XOR encryption scheme, we derive the SER for both schemes. In the XOR encryption scheme, the channel coefficients are

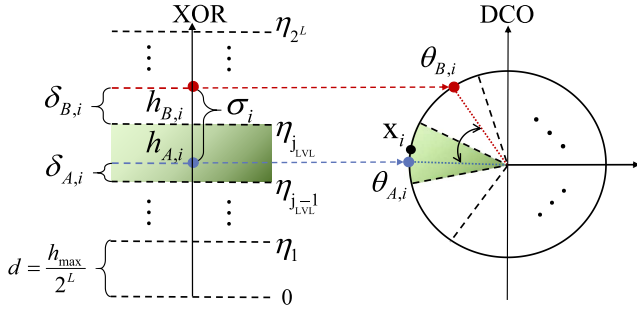


Fig. 6. Schematic of SER calculation for the schemes of XOR and DCO.

quantized into key bits, denoted as $\mathbf{q}_A$ and $\mathbf{q}_B$. For the convenience of encryption, we assume that the number of quantization levels is the same as the modulation order. The quantization rule for the i th symbol $\hat{h}_{u,i}$ can be expressed as

$$\mathbf{q}_{u,i} = \begin{cases} q_{2^L}, & \hat{h}_{u,i} \in [\eta_{2^L-1}, \eta_{2^L}], \\ \vdots & \vdots \\ q_{j_{LVL}}, & \hat{h}_{u,i} \in [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}], \\ \vdots & \vdots \\ q_1, & \hat{h}_{u,i} \in [0, \eta_1] \end{cases} \quad (29)$$

where $u \in \{A, B\}$; $\hat{h}_{u,i}$ denotes the real or image part of the estimated channel coefficient of the i th subcarrier at user u , and $\eta_{j_{LVL}-1} = \hat{h}_u^{\max}((j_{LVL} - 1)/2^L)$ denotes the j_{LVL} th quantization level. Therefore, the SER of the XOR scheme can be expressed as the sum of errors that occur at each range

$$\begin{aligned} Pr^{\text{XOR}} &= P(q_{A,i} \neq q_{B,i}) \\ &= P(\hat{h}_{A,i} \in [\eta_0, \eta_1]) \\ &\quad \times P(\hat{h}_{A,i} \in [\eta_0, \eta_1] \cap \hat{h}_{B,i} \notin [\eta_0, \eta_1]) \\ &\quad + \sum_{j_{LVL}=2}^{N-1} P(\hat{h}_{A,i} \in [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}]) \\ &\quad \times P(\hat{h}_{A,i} \in [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}] \cap \hat{h}_{B,i} \notin [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}]) \\ &\quad + P(\hat{h}_{A,i} \in [\eta_{N-1}, \eta_N]) \\ &\quad \times P(\hat{h}_{A,i} \in [\eta_{N-1}, \eta_N] \cap \hat{h}_{B,i} \notin [\eta_{N-1}, \eta_N]) \end{aligned} \quad (30)$$

where j_{LVL} denotes the quantization interval index.

As shown in Fig. 6, we define quantization error $\delta_{u,i}$ as the difference between the channel measurement value and its quantization level, which is given by

$$\delta_{u,i} = \hat{h}_{u,i} - \eta_{j_{LVL}-1} \quad (31)$$

Furthermore, we define the error between the uplink and downlink channel estimation results as

$$\sigma_i = \hat{h}_{B,i} - \hat{h}_{A,i} = z_{B,i} - z_{A,i} \quad (32)$$

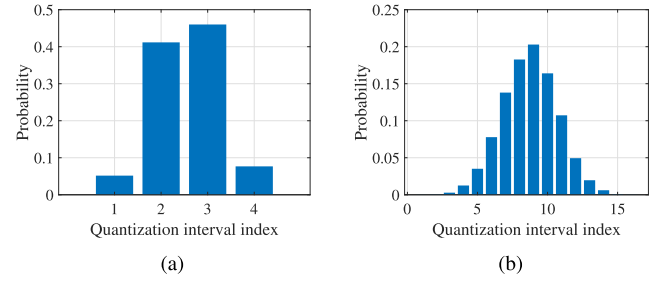


Fig. 7. Probability distributions of quantized real part of $\hat{h}_{u,i}$: (a) $L = 2$ and (b) $L = 4$.

where σ_i follows a complex normal distribution as $\sigma_i \sim \mathcal{CN}(0, 2\sigma_n^2/P)$. Thus, we can obtain:

$$\begin{aligned} P(\hat{h}_{A,i} \in [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}] \cap \hat{h}_{B,i} \notin [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}]) &= \\ \begin{cases} P(\sigma_i > d - \delta_{A,i}), & \text{if } j_{LVL} = 1, \\ P(\sigma_i > d - \delta_{A,i}) + P(\sigma_i < -\delta_{A,i}), & \text{if } 2 \leq j_{LVL} \leq N-1, \\ P(\sigma_i < -\delta_{A,i}), & \text{if } j_{LVL} = N \end{cases} \end{aligned} \quad (33)$$

where the cases $j_{LVL} = 1$ and $j_{LVL} = N$ correspond to $\hat{h}_{A,i}$ falling within the edge quantization intervals.

As illustrated in Fig. 7, the probability distributions of the quantized real parts of channel coefficients $\hat{h}_{u,i}$ are displayed, based on a total number of 10 000 samples from the quantization process. Specifically, Fig. 7(a) corresponds to $L = 2$ with four quantization intervals, which is equivalent to QPSK, and Fig. 7(b) corresponds to $L = 4$ with 16 quantization intervals, which is equivalent to 16QAM. It can be seen that most of the real values of $\hat{h}_{u,i}$ are concentrated near the central intervals, with significantly fewer values lying in the edge intervals. This phenomenon is attributed to the Gaussian distribution of $\hat{h}_{u,i}$, where the probability density peaks at the mean and exponentially decreases toward both tails. Consequently, the probabilities associated with the edge intervals are negligible, and the SER of the XOR scheme can be expressed as

$$\begin{aligned} Pr^{\text{XOR}} &= P(\hat{h}_{A,i} \in [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}] \cap \hat{h}_{B,i} \notin [\eta_{j_{LVL}-1}, \eta_{j_{LVL}}]) \\ &= P(\sigma_i > d - \delta_{A,i}) + P(\sigma_i < -\delta_{A,i}). \end{aligned} \quad (34)$$

Here, d represents the quantization interval.

In contrast, in the DCO encryption scheme, the quantization step is omitted, and the channel measurements are taken as a remainder, which are then mapped within range $[0, 2\pi)$. To compare the performance of the DCO scheme with the quantization scheme under the identical conditions, we assume that there are also 2^L phase angles of modulated symbols. Consequently, the SER of the DCO scheme can be expressed as follows:

$$\begin{aligned} Pr^{\text{DCO}} &= P\left(|\theta_A - \theta_B| > \frac{2\pi}{2 \cdot 2^L}\right) \\ &= P\left(2\pi \frac{|\hat{h}_{A,i} - \hat{h}_{B,i}|}{h_{\max}} > \frac{\pi}{2^L}\right) \\ &= P\left(|\sigma_i| > \frac{d}{2}\right). \end{aligned} \quad (35)$$

Here, we denote the probability density function (PDF) and cumulative distribution function (CDF) of δ as $f_\delta(x)$ and $F_\delta(x)$. Similarly, we represent the PDF and CDF of σ_i as $f_\sigma(x)$ and $F_\sigma(x)$, respectively. Equations (34) and (35) can thus be rewritten in terms of those PDFs and CDFs as

$$\begin{aligned} P_r^{\text{XOR}} &= P(\sigma_i > d - \delta_i) + P(\sigma_i < -\delta_i) \\ &= 1 - P(-\delta_i < \sigma_i < d - \delta_i) \\ &= 1 - \int_0^d f_\delta(x) \times (F_\sigma(d - x) - F_\sigma(-x)) dx \\ &= 1 - \int_0^d (F_\sigma(d - x) - F_\sigma(-x)) dF_\delta(x) \\ &= 1 - (F_\sigma(d - x) - F_\sigma(-x)) \times F_\delta(x) \Big|_0^d \\ &\quad + \int_0^d F_\delta(x) d(F_\sigma(d - x) - F_\sigma(-x)) \quad (36) \end{aligned}$$

and

$$\begin{aligned} P_r^{\text{DCO}} &= P\left(|\sigma_i| > \frac{d}{2}\right) = 1 - \left(F_\sigma\left(\frac{d}{2}\right) - F_\sigma\left(-\frac{d}{2}\right)\right) \\ &= 1 - \frac{1}{2} - \frac{1}{2} \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) + \frac{1}{2} - \frac{1}{2} \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) \\ &= 1 - \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right). \quad (37) \end{aligned}$$

Letting $\int_0^d F_\delta(x) d(F_\sigma(d - x) - F_\sigma(-x)) = K(d)$, we can derive the following expression for the difference as

$$\begin{aligned} P_r^{\text{DCO}} - P_r^{\text{XOR}} &= (F_\sigma(0) - F_\sigma(-d)) \cdot F_\delta(d) - K(d) - \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) \\ &= \frac{1}{2} - F_\sigma(-d) - K(d) - \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) \\ &= \frac{1}{2} - \left(\frac{1}{2} - \frac{1}{2} \operatorname{erf}\left(\frac{d}{\sqrt{2}\sigma_n}\right)\right) - K(d) - \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) \\ &= \frac{1}{2} \operatorname{erf}\left(\frac{d}{\sqrt{2}\sigma_n}\right) - K(d) - \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right). \quad (38) \end{aligned}$$

Since $F_\sigma(x) \geq 0$, $F_\delta(x)$ is an increasing function and $d - x > -x$, we can derive $K(d) > 0$. Additionally, as the error function $\operatorname{erf}(x)$ is convex for $x > 0$ and $\operatorname{erf}(0) = 0$, we have

$$\operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) > \frac{1}{2} \operatorname{erf}\left(\frac{d}{\sqrt{2}\sigma_n}\right). \quad (39)$$

From the above derivation, we can deduce the following inequality regarding the performance difference:

$$\begin{aligned} P_r^{\text{DCO}} - P_r^{\text{XOR}} &= \frac{1}{2} \operatorname{erf}\left(\frac{d}{\sqrt{2}\sigma_n}\right) - K(d) - \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) \\ &< \frac{1}{2} \operatorname{erf}\left(\frac{d}{\sqrt{2}\sigma_n}\right) - \operatorname{erf}\left(\frac{d}{2\sqrt{2}\sigma_n}\right) < 0. \quad (40) \end{aligned}$$

Equation (40) proves that our proposed DCO scheme has better SER performance compared to the XOR scheme.

B. Security Analysis

1) *Information-Theoretic Secrecy*: In this section, we adopt the secrecy capacity as the performance metric to examine

whether the DCO scheme can achieve better information-theoretic secrecy than the traditional XOR scheme. Normally, the secrecy capacity, denoted as C_s , can be defined as [34]

$$C_s = C_m - C_w \quad (41)$$

where C_m and C_w denote the Shannon capacities of the legitimate and eavesdropping channels, respectively. As DCO focuses on real-time wireless communications, the secrecy capacity for one realization (γ_m, γ_w) of the quasi-static complex fading wiretap channel is given by

$$C_s = \begin{cases} \log_2(1 + \gamma_m) - \log_2(1 + \gamma_w), & \text{if } \gamma_m \geq \gamma_w \\ 0, & \text{if } \gamma_m < \gamma_w \end{cases} \quad (42)$$

where γ_m and γ_w are the SNRs of the received signals propagating through the legitimate and eavesdropping channels, respectively. After encryption, the received signal at Eve can be represented as

$$\begin{aligned} y_w(i) &= h_w(i)x(i)(\cos \theta_i + j \sin \theta_i) + z_w(i) \\ &= h_w(i)x(i)\left(1 - 2 \sin^2 \frac{\theta_i}{2} + j \sin \theta_i\right) + z_w(i). \quad (43) \end{aligned}$$

On the other hand, Bob, being the intended recipient who possesses the decryption patterns, can decrypt the received signal. The received signal at Bob can be represented as

$$y_m(i) = h_m(i)x(i) + z_m(i). \quad (44)$$

Therefore, instantaneous SNRs $\gamma_m(i)$ and $\gamma_w(i)$ can be represented as

$$\gamma_m(i) = \frac{P|h_m(i)|^2}{|z_m(i)|^2} = \frac{P|h_m|^2}{|z_m|^2} = r_m \quad (45)$$

and

$$\begin{aligned} \gamma_w(i) &= \frac{P|h_w(i)|^2}{z_w + P \cdot |h_w(i)|^2 |j \sin \theta_i - 2 \sin^2 \frac{\theta_i}{2}|^2} \\ &= \frac{P|h_w|^2}{z_w + 4P|h_w|^2 \sin^2 \frac{\theta_i}{2}} = \frac{r_w}{1 + 4r_w \sin^2 \frac{\theta_i}{2}} \quad (46) \end{aligned}$$

where $r_m = (P|h_m|^2/|z_m|^2)$ and $r_w = (P|h_w|^2/|z_w|^2)$ are the original SNRs of the unencrypted legitimate channel and wiretap channel, respectively.

In both DCO and XOR schemes, the legitimate receiver can correctly decrypt the symbols, and, hence, the average SNR of the legitimate channels can be written as

$$\bar{\gamma}_m^{\text{DCO}} = \bar{\gamma}_m^{\text{XOR}} = \bar{\gamma}_m = \mathbb{E}\{r_m\}. \quad (47)$$

In the DCO scheme, θ_i is distributed abiding by a uniform distribution over $[0, 2\pi)$, based on which the average SNR of the eavesdropping channel can be written as

$$\begin{aligned} \bar{\gamma}_w^{\text{DCO}} &= \mathbb{E}\left\{\frac{r_w}{1 + 4r_w \sin^2 \frac{\theta_i}{2}}\right\} \\ &= \frac{r_w}{2\pi} \int_0^{2\pi} \frac{1}{1 + 4r_w \sin^2 \frac{\theta}{2}} d\theta = \frac{r_w}{\sqrt{1 + 4r_w}}. \quad (48) \end{aligned}$$

While in the XOR scheme, the bit-level encryption can be equated to a rotationally fixed angle, e.g., in QPSK, $0, 1\pi/2$,

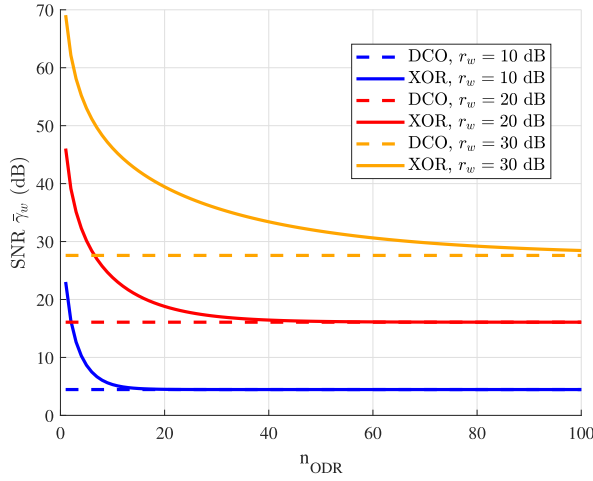


Fig. 8. Average SNR of eavesdropping channel versus modulation order n_{ODR} .

π , or $3\pi/2$. As a result, the average SNR of the eavesdropping channel can be written as

$$\bar{\gamma}_w^{xor} = \frac{1}{n_{ODR}} \sum_{k=0}^{n_{ODR}-1} \frac{r_w}{1 + 4r_w \sin^2 \frac{\pi k}{n_{ODR}}} \quad (49)$$

where n_{ODR} represents the modulation order, and k is the index of discrete phase values ranging from 0 to $n_{ODR} - 1$.

As $n_{ODR} \rightarrow \infty$, these discrete points will become considerably dense, and $(\pi k/n_{ODR})$ will thereby approximate uniform sampling over the interval $[0, \pi]$. When $n_{ODR} \rightarrow \infty$, $(\pi k/n_{ODR}) \in [0, \pi]$, let $\theta \in [0, 2\pi]$, and we have $(\pi k/n_{ODR}) = (\theta/2)$. Thus, we can have

$$\bar{\gamma}_w^{XOR} \approx \frac{r_w}{2\pi} \int_0^{2\pi} \frac{1}{1 + 4r_w \sin^2 \frac{\theta}{2}} d\theta. \quad (50)$$

The form of this integral matches exactly with $\bar{\gamma}_w^{DCO}$. Therefore, as $n_{ODR} \rightarrow \infty$, we would expect that $\bar{\gamma}_w^{XOR} \rightarrow \bar{\gamma}_w^{DCO}$.

By numerically evaluating (48)–(50), as shown in Fig. 8, it is observed that $\bar{\gamma}_w^{XOR}$ decreases monotonically. When $n_{ODR} = 1$, $\bar{\gamma}_w^{XOR} = r_w$, which is greater than $\bar{\gamma}_w^{DCO}$. Therefore, we can deduce

$$\bar{\gamma}_w^{XOR} > \bar{\gamma}_w^{DCO}. \quad (51)$$

As shown in this figure, the average SNR for DCO remains constant irrespective of n_{ODR} , which is in line with our expectations. On the other hand, the average SNR for XOR gradually approaches that of DCO as n_{ODR} increases, also consistent with our expectations. These results validate our theoretical analysis and demonstrate that XOR can provide comparable performance to DCO under the same wiretap channel condition.

Therefore, we can obtain

$$\begin{aligned} C_s^{DCO} - C_s^{XOR} &= \log\left(\frac{1 + \bar{\gamma}_m^{DCO}}{1 + \bar{\gamma}_w^{DCO}}\right) - \log\left(\frac{1 + \bar{\gamma}_m^{XOR}}{1 + \bar{\gamma}_w^{XOR}}\right) \\ &= \log\left(\frac{1 + \bar{\gamma}_m^{DCO}}{1 + \bar{\gamma}_m^{XOR}}\right) > 0. \end{aligned} \quad (52)$$

This demonstrates that our proposed scheme can achieve superior secrecy capacity at the same SNR due to the uniformly distributed dynamic constellation rotation, which prevents the signals from being overheard through the eavesdropping channel.

2) *Secrecy of Initial Key and Data Frames:* For the generation of the initial key, a private pilot with random phases is employed to generate similar initial constellation patterns. The eavesdropper can only obtain the observed results of $\varphi_{A,i} + \theta_{m,i}(0)$ and $\varphi_{B,i} + \theta_{m,i}(0)$, but is not able to deduce $\varphi_{A,i} + \theta_{m,i}(0) + \varphi_{B,i}$, thus failing to obtain the information regarding the initial phase.

For data frames, an attacker may try to recover the data content or launch a traffic analysis attack to obtain information, such as communication patterns and modulation schemes. By our scheme, constellation patterns are updated dynamically and are accessible only to the legitimate ends. Hence, the attacker cannot identify the constellation and carry out proper decryption. Even if the attacker can obtain part of the transmitted data and generate a closer initial value to x_0 , the constellations decrypted by the attacker are randomly distributed due to the introduction of chaotic sequences.

3) *Defense Against Symbol Detection:* As the DCO encryption scheme obscures the location of the original data symbols in the constellation diagram, an eavesdropper may attempt to employ traditional modulation identification methods, such as automatic modulation classification (AMC) or deterministic modulation classification (DMC) for symbol detection. AMC operates based on cyclic features, which means that different digital modulation schemes possess distinct time-dependent periodic information that can be exploited by attackers. In contrast, DMC consistently utilizes the standard constellation shape as its foundation for determining the modulation scheme of received symbols. Employing this algorithm, the eavesdropper can construct a scatter constellation graph of the received symbols and employs fuzzy c-mean clustering to recover a robust constellation graph.

Fig. 9 presents a comparison of standard and DCO encryption constellation maps at an SNR of 15 dB. It is evident that following DCO encryption and noise interference, the arrangement of constellation points is entirely disrupted and uniformly distributed across the constellation space. DCO does not perform similarly to any known digital modulation technique, rendering both AMC and DMC methods unable to identify the modulation scheme of the encrypted symbols. Consequently, the eavesdropper cannot identify and match the received constellation diagram, preventing the recovery of the original symbol. On the other hand, the legitimate receiver can use the highly symmetrical decryption sequence to perform reverse rotation decryption operation, ensuring accurate identification and matching of the constellation diagram.

In addition, an adversary may also attempt to exploit the statistical changes in the encrypted OFDM symbols to determine whether the key can be disclosed from the constellation map. It is widely known that a chaotic sequence generated by a chaotic system exhibits random behaviors and remains unpredictable for an extended period. The process of constellation rotation is equivalent to concealing the symbol phase, thereby eliminating

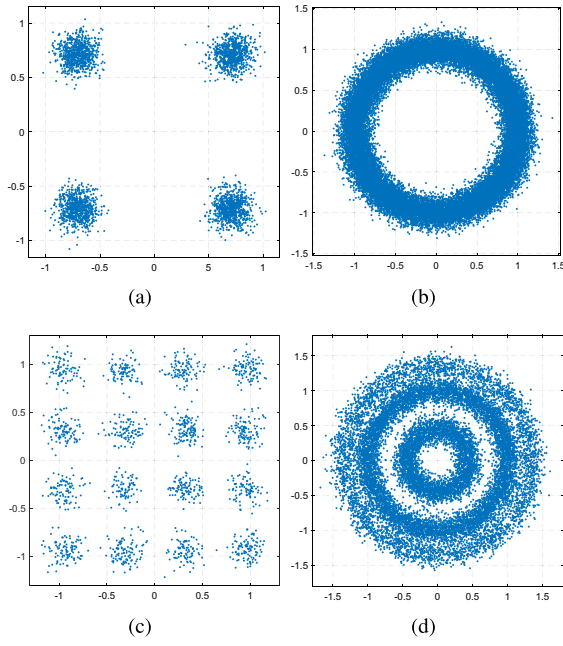


Fig. 9. Comparison of constellation maps (SNR = 15 dB): (a) standard QPSK constellation; (b) DCO encryption QPSK constellation; (c) standard 16-QAM constellation; and (d) DCO encryption 16-QAM constellation.

the phase characteristics associated with different modulation schemes. As depicted in Fig. 10(a), the encrypted OFDM symbols result in a PDF close to a uniform distribution, which confirms a high degree of mixing and distribution of encrypted OFDM symbols across the entire space. Nevertheless, the legitimate receiving end can utilize a highly consistent phase sequence to reversely rotate the encrypted symbols, ultimately obtaining the OFDM symbol with modulation characteristics, as illustrated in Fig. 10(b).

4) *Defense Against Differential Attack*: A differential attacker may exploit the relation between the results of two encrypted symbols to retrieve the relation between the initial symbols. When the traditional XOR scheme is applied in a slowly varying environment, an eavesdropper can launch a differential attack by performing the XOR operation on the ciphertext with highly consistent keys generated from adjacent frames

$$\begin{aligned} \mathbf{x}^c(k) \oplus \mathbf{x}^c(k+1) &= (\mathbf{q}_A(k) \oplus \mathbf{x}(k)) \oplus (\mathbf{q}_A(k) \oplus \mathbf{x}(k+1)) \\ &= \mathbf{x}(k) \oplus \mathbf{x}(k+1). \end{aligned} \quad (53)$$

In contrast, the two adjacent encrypted OFDM symbols in our proposed DCO scheme are given by

$$\mathbf{x}^c(k) = E(\mathbf{e}^A(k), \mathbf{x}(k)) = E(\theta^A(k) + \mathbf{c}^A(k), \mathbf{x}(k)) \quad (54)$$

and

$$\begin{aligned} \mathbf{x}^c(k+1) &= E(\mathbf{e}^A(k+1), \mathbf{x}(k+1)) \\ &= E(\theta^A(k+1) + \mathbf{c}^A(k+1), \mathbf{x}(k+1)) \end{aligned} \quad (55)$$

where $\mathbf{c}^A(k)$ and $\mathbf{c}^A(k+1)$ are generated from the transmitted data of different frames by a one-way chaotic function, and

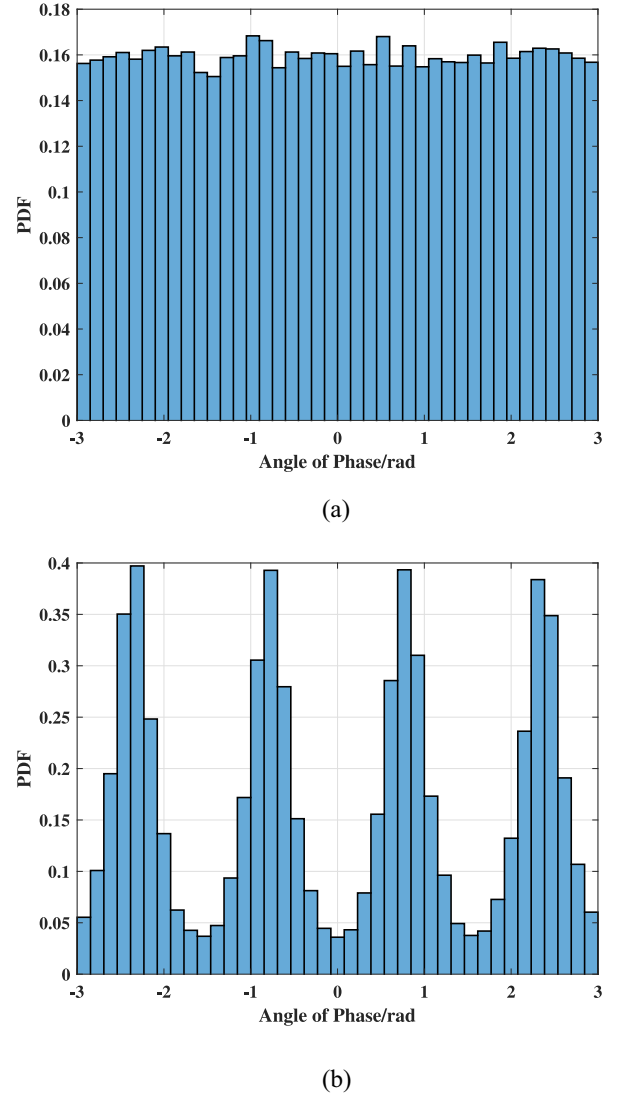


Fig. 10. PDF of the angle of received OFDM symbols. (a) Encrypted symbols. (b) Decrypted symbols.

$\theta^A(k+1)$ and $\theta^A(k)$ are the channel estimation results. It is rather difficult by nature for an attacker to determine the alteration rate of constellation patterns. Besides, due to the sensitivity of chaotic sequences, the differences between adjacent frames can be significant. Consequently, differential attacks become ineffective when the proposed DCO scheme is in use.

5) *Defense Against Injection Attack*: In the *Initial constellation pattern* phase, Eve simultaneously injects her random phase $\varphi_{E,i}$ into both Alice and Bob, aiming to disrupt their initial condition generation process. This injection attack essentially alters the phase of the channel matrices between Alice and Bob; therefore, after phase compensation, the phase of the nonnoise term extracted by Alice becomes

$$\varphi_{A,i} + \theta_{m,i}(0) + \varphi_{E,i} + \varphi_{B,i}, \quad (56)$$

and the phase extracted by Bob becomes

$$\varphi_{B,i} + \theta_{m,i}(0) + \varphi_{E,i} + \varphi_{A,i}. \quad (57)$$

At this point, the phase injected by Eve can be considered part of the channel phase, so Alice and Bob can still obtain the same initial patterns, denoted as Φ , while Eve cannot obtain the initial patterns. Under this condition, Eve's injection attack does not affect the security of the DCO scheme.

6) *Defense Against Channel Replay Attack*: In a channel replay attack, Eve has access to channel information $\theta_B(k) \approx \theta_A(k)$ that is similar and reciprocal to Bob. Eve tries to decrypt the encrypted symbol $\mathbf{x}^c(k)$ with the known channel information

$$\begin{aligned}\hat{\mathbf{x}}_E(k) &= \text{diag}(e^{-j\theta_{B,0}(k)}, \dots, e^{-j\theta_{B,N-1}(k)}) \mathbf{x}^c(k) \\ &= \text{diag}(e^{-j\theta_{A,0}(k)}, \dots, e^{-j\theta_{A,N-1}(k)}) \mathbf{E}^A(k) \mathbf{x}(k)\end{aligned}\quad (58)$$

where

$$\begin{aligned}\text{diag}(e^{-j\theta_{A,0}(k)}, \dots, e^{-j\theta_{A,N-1}(k)}) \mathbf{E}^A(k) \\ = \text{diag}(e^{j c_{0,I}(k)}, \dots, e^{j c_{N-1,I}(k)}).\end{aligned}\quad (59)$$

It can be seen that although the phase impact of the channel is eliminated by the same channel, the impact of the random phase of the data $c_{i,I}(k)$ is still retained in the data decrypted by the attack. Therefore, the channel replay impact is ineffective for this scheme.

7) *Defense Against Message Manipulating Attack*: Against the threat that an attacker may force a legitimate terminal to send duplicate data frames (e.g., fixed command response, periodic status reports) through protocol vulnerabilities or traffic manipulation, the DCO scheme still ensures security. First, the key generation of DCO relies not only on data randomness, but also on the intrinsic randomness of the channel. Even though the content of data frames may be repetitive, CSI remains random due to environmental perturbations, and thus generated keys remain independent. In addition, in order to cope with the extreme case where the channel changes slowly and the data is repeated, DCO employs a flexible sliding-window mechanism and an entropy-boosting scheme, which ensures the reliability of the key's random source.

C. Practical Challenge

Although both encryption schemes provide strong security, the real time requirements for dynamic initial condition and phase updates become more stringent. Let n_{BIT} denote the number of processing units. For the XOR scheme n_{BIT} is the number of plaintext bits, and for the DCO scheme with QPSK modulation n_{BIT} is the number of symbols, namely, the bit count divided by four. Define n_{TM1} as the number of operations per unit for generating the encryption key or phase, and n_{TM2} as the number of operations per unit for performing encryption. The total computational cost can thus be written as

$$O(n_{\text{TM1}} n_{\text{BIT}} + n_{\text{TM2}} n_{\text{BIT}}) \quad (60)$$

hence both schemes exhibit linear time complexity $O(n_{\text{BIT}})$. Moreover, since the logistic map is a 1-D discrete chaotic system that does not require numerical integration of differential equations, its constant-factor overhead remains small, making it especially suitable for hardware-constrained physical-layer security implementations.

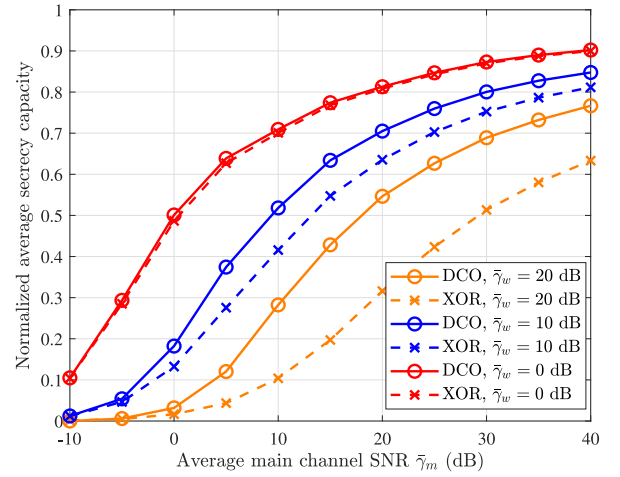


Fig. 11. Normalized average secrecy capacity versus $\bar{\gamma}_m$.

However, achieving this theoretical efficiency in real-time wireless deployments also hinges on robust frame synchronization and precise symbol timing synchronization to ensure that encrypted content is correctly framed and demodulated. First, frame synchronization requires encrypting only the data payload and preserving in plaintext the preamble sequences used to detect the start of a frame [35], [36]. Second, symbol timing synchronization requires precisely aligning the start of each OFDM symbol within a frame so that the FFT window coincides with the symbol boundaries. Common methods include CP based autocorrelation detection and the Schmidl Cox algorithm, both of which estimate the symbol start position while simultaneously compensating for frequency offset [37].

VI. SIMULATION RESULTS

In this section, simulation results are shown and discussed for OFDM systems using our proposed encryption schemes to verify the security and reliability analyzes given in the previous section. The simulation parameters are adopted as per the IEEE 802.11a standard, and the number of iterations of the logistic chaotic sequence is set to be 100.

Fig. 11 compares the normalized average secrecy capacity of the DCO and XOR schemes in a quasi-static fading channel. Normalization is performed with respect to the capacity of an AWGN channel with the SNR equal to $\bar{\gamma}_m$. It can be seen from this figure that the secrecy capacity increases with the increase of the SNR of the legitimate channel. Furthermore, the average secrecy capacity of the DCO scheme is higher than that of the XOR scheme, consistent with the theoretical derivation given in (52). In addition, the larger the SNR of the eavesdropper is, the larger the secrecy capacity gain brought by the DCO scheme relative to the XOR scheme will be. When $\bar{\gamma}_w = 10$ dB, there is a secrecy capacity gain of 0.1 bps. This suggests that DCO encryption can effectively degrade the eavesdropping capability and increase secrecy capacity, even when the eavesdropper possesses favorable channel conditions.

The secrecy capacity of DCO can also be characterized in terms of outage probability. The outage probability is

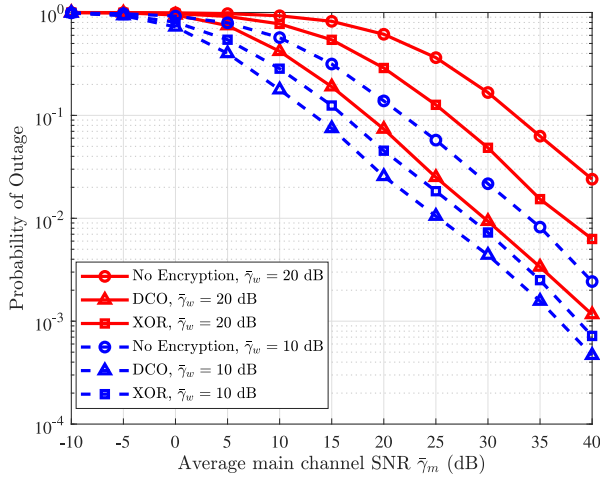
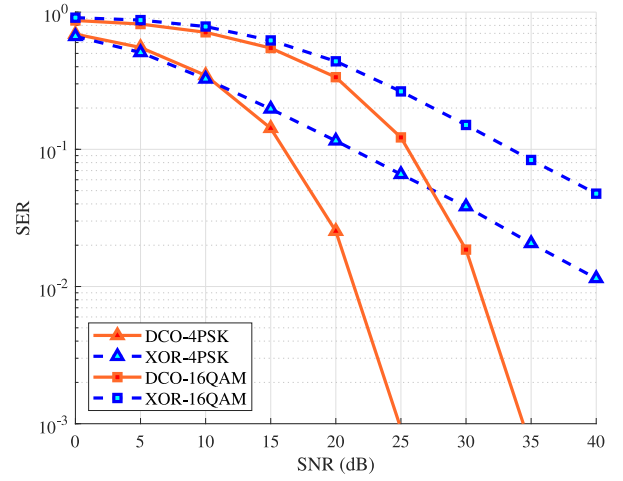

 Fig. 12. Outage probability versus $\bar{\gamma}_m$, given $R_s = 0.1$.


Fig. 14. SER of different encryption schemes.

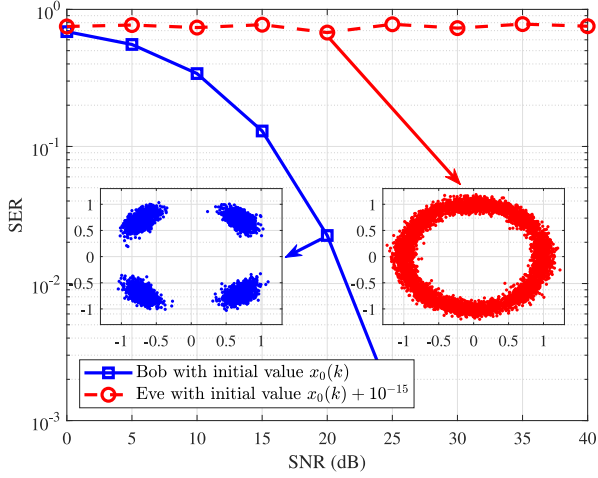


Fig. 13. SER of Bob and Eve under different initial values for QPSK.

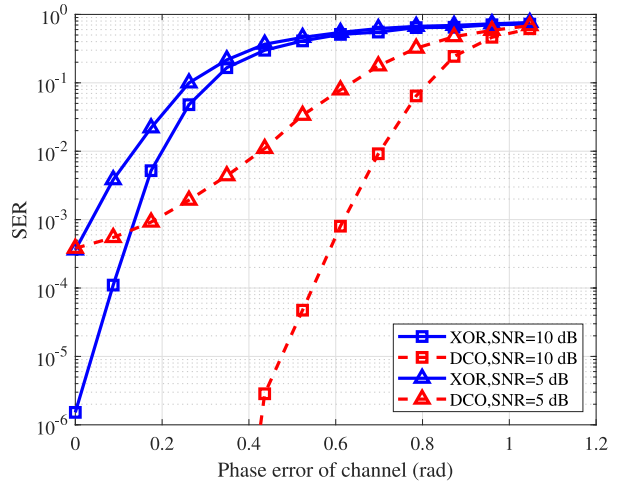


Fig. 15. SER of different encryption schemes versus phase error with different SNRs.

formulated as $\mathcal{P}_{\text{out}}(R_s) = P[C_s < R_s]$, which represents the probability of the instantaneous secrecy capacity being below a target secrecy rate R_s . Fig. 12 shows the outage probability versus $\bar{\gamma}_m$, for selected values of $\bar{\gamma}_w$ and a normalized target secrecy rate equal to 0.1. We can observe that the outage probability monotonically decreases with respect to $\bar{\gamma}_m$ but monotonically increases with respect to $\bar{\gamma}_w$. Moreover, if $\bar{\gamma}_m \ll \bar{\gamma}_w$, the outage probability approaches one. The outage probability of the DCO scheme is higher than that of the XOR scheme, and a larger $\bar{\gamma}_w$, leads to a greater SNR, with a gain of about 6.5 dB at the outage probability of 0.1 ($\bar{\gamma}_w = 20$ dB). The above simulation results demonstrate the superior security performance of the proposed DCO scheme compared to the XOR scheme.

Fig. 13 exhibits the SER performance and received constellations of Bob and Eve with slightly different initial values. It can be seen that after constellation rotation encryption and noise interference, the arrangement of constellation patterns is completely disrupted and evenly distributed in the constellation space. For the legitimate receiver with $x_k^B(0)$, the OFDM symbols can be correctly decrypted after transmission, while the illegal receiver with a slight difference of 10^{-15}

ends up with an SER of 0.7, indicating that the illegal user is incapable of identifying the received constellation patterns and recovering any effective information. Therefore, the security advantage brought by the key sensitivity of DCO is verified.

In Fig. 14, the SER performance of our proposed dynamic phase encryption scheme is simulated and compared with the conventional XOR encryption scheme. As the simulation results shown in this figure, we can find that as the SNR increases, the SER corresponding to different schemes gradually decreases. Compared with the XOR encryption scheme, our proposed scheme achieves better SER performance, where at the SER of 0.1, a 4.5 dB gain can be achieved when using the QPSK modulation. This is because quantization will cause the loss of part of the transmitted information, leading to an increased disagreement between the encrypted and decrypted sequences, and ultimately resulting in the degradation of SER performance. This result agrees with the theoretical analysis of SER given in the previous section.

In Fig. 15, we investigate the robustness performance of the proposed scheme under different phase errors at SNR = 5 dB and 10 dB. The phase error is uniformly distributed in

$[0, 2\pi)$, and is increased by $\pi/36$ interval from 0 to $\pi/3$, which is modeled according to (19). It can be seen that the SER performance gradually decreases as the phase error increases and eventually approaches about 0.7 at around $\pi/3$. This is because the channel phase error of the system can lead to symbol misclassification, and when the channel phase estimation error is greater than $\pi/3$, the symbol is completely misclassified. In addition, as the order of modulation increases, the phase robustness improves significantly, tolerating a larger phase error while maintaining the same SER performance. Furthermore, when the channel phase error is less than $\pi/12$, the phase error has less impact on the SER of the proposed analog-based DCO scheme, and the SER curve is relatively flat, while the fluctuation of the digital-based XOR scheme is larger, and the corresponding SER rises rapidly. This indicates that the DCO scheme has a certain tolerance to the channel phase estimation error and is more robust in the case of channel phase error.

These simulation results are consistent with the theoretical analysis, which verify the security and reliability advantages of the proposed DCO scheme.

VII. CONCLUSION

In this article, we proposed the DCO scheme to enhance the security of wireless transmissions. The DCO scheme neither relies on quasi-static channel assumption nor requires a trusted third party for key distribution. Furthermore, it combines the randomness of previously transmitted data and initial channel information to generate adaptive constellation patterns, providing enhanced confidentiality protection against various attacks. Specifically, the dynamic obfuscation of constellation maps at the sender prevents eavesdroppers from identifying and matching constellation templates, thereby securing the transmitted data. We further developed mathematical models for information security and SER analysis of DCO, demonstrating its superior information-theoretic confidentiality and transmission reliability compared to XOR encryption. The simulation results corroborated our theoretical derivations and indicated that the DCO scheme can effectively prevent eavesdroppers from obtaining useful information and exhibits superior SER, secrecy capacity, and greater robustness against phase errors compared to the traditional XOR encryption scheme. The proposed scheme establishes a theoretical framework for dynamic physical-layer security. Future research will focus on practical implementation aspects, including synchronization mechanisms and real-time processing optimization, to facilitate deployment in real-world system.

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